

ECE0202: Embedded Systems and Interfacing

Lab 4: Keypad Scanning (in assembly)

Due: TBD

Objectives

- Understand I/O multiplexing technique
- Be familiar with keypad scanning algorithms
- Implement software debouncing

Pre-Lab Reading

- Chapter 14.9 Keypad Scanning

Deliverables – total 100 points

- (40 points) – demonstration of keypad scanning that displays the pressed character on the Tera Term.
- (35 points) – Code submission. Code should use software debouncing and be well-written and commented.
- (25 points) Submission of the pre-lab register tables and post-lab questions. Indicate how each group member participates and contributes to the lab at the end of the lab report.
- Indicate each group member's Participation and Contribution.

Please submit your code and pre-lab answers and your Circuit photo as a pdf to Black board.

Keyboard Interface

The 4x4 keypad used in this lab requires 8 pins (4 row pins and 4 column pins). In this lab, the connection between the keypad and the discovery kit is shown in the following table.

Row	R1 → PC0	R2 → PC1	R3 → PC2	R4 → PC3
Column	C1 → PB1	C2 → PB2	C3 → PB3	C4 → PB5

All pins of the input port (C1, C2, C3, and C4) need be pulled up to 3.3V with 2.2kΩ resistors **on breadboard**; Within the processor, each GPIO pin can be pulled up via an internal resistor (between 20 and 55kΩ), but the internal pull-up current capability is too weak, and therefore an external pull-up resistor is required, as drawn in Fig. 1.

When looking at the front side of the keypad, the pins on the connector from left to right are:

R1 – R2 – R3 – R4 – C1 – C2 – C3 – C4

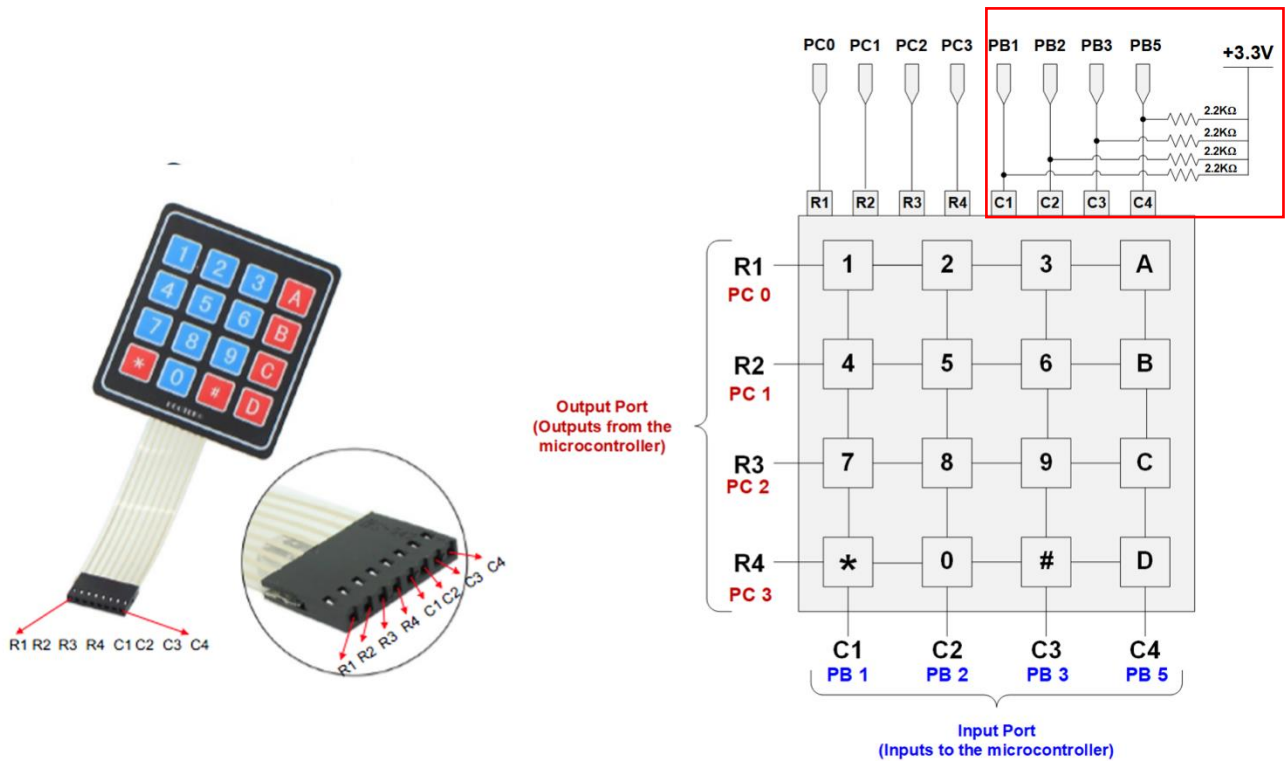


Figure 1- Picture and schematic of the keypad

The maximum current a GPIO pin can source or sink is 20 mA. When calculating the value of external pull-up resistors, make sure that the current does not exceed 20 mA. For example:

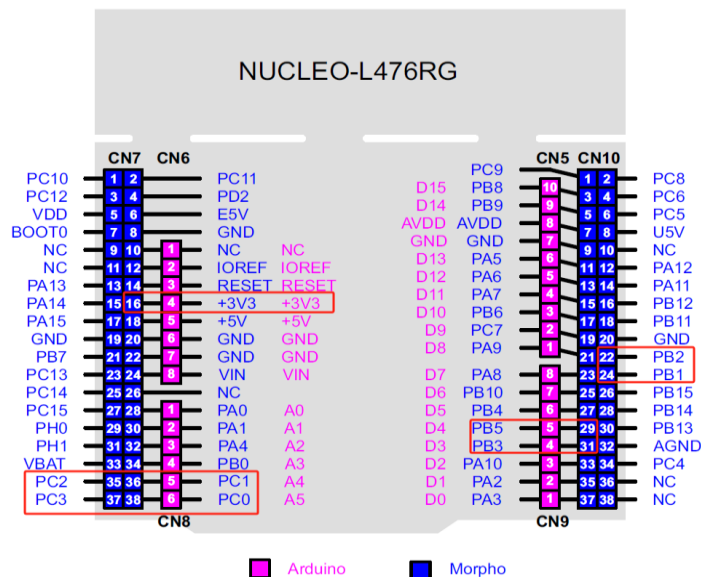
$$\frac{3V}{2.2k\Omega} = 1.4mA$$

Dupont wire



Please tear the Dupont wire, do not take a whole at a time, save a little for other students

After use, return the three types of wires in separate categories



Arduino Morpho

On the Nucleo board, all pins in the input port (**PB1, PB2, PB3, and PB5**) are connected to ground via a 100 nF capacitor. A very short delay should be added before reading the input port, as seen later in the software flowchart in Fig. 3.

ASCII

In order to write to the Tera Term, you must store the code for an ASCII character into a memory location. In lab 1, the string to be displayed on the Tera Term is stored at a memory location called “str”. This can also be used as the memory location that stores the character you display in this lab.

To display a character on the Tera Term, you must store the associated ASCII code in the memory location “str”, load the memory address of “str” into r0, and then run the instruction *BL USART2_Write*. The following table gives ASCII codes for many characters. Note that these are in decimal!

ASCII value	Character	ASCII value	Character	ASCII value	Character
000	^@	043	+	086	V
001	^A	044	,	087	W
002	^B	045	-	088	X
003	^C	046	.	089	Y
004	^D	047	/	090	Z
005	^E	048	0	091	[
006	^F	049	1	092	\
007	^G	050	2	093	]
008	^H	051	3	094	^
009	^I	052	4	095	_
010	^J	053	5	096	'
011	^K	054	6	097	a
012	^L	055	7	098	b
013	^M	056	8	099	c
014	^N	057	9	100	d
015	^O	158	:	101	e
016	^P	059	;	102	f
017	^Q	060	<	103	g
018	^R	061	=	104	h
019	^S	062	>	105	i
020	^T	063	?	106	j
021	^U	064	@	107	k
022	^V	065	A	108	l
023	^W	066	B	109	m
024	^X	067	C	110	n
025	^Y	068	D	111	o
026	^Z	069	E	112	p
027	^[	070	F	113	q
028	^\	071	G	114	r
029	]`	072	H	115	s
030	^^	073	I	116	t
031	^-	074	J	117	u
032	[space]	075	K	118	v
033	!	076	L	119	w
034	"	077	M	120	x
035	#	078	N	121	y
036	\$	079	O	122	z
037	%	080	P	123	{
038	&	081	Q	124	
039	'	082	R	125	}
040	(	083	S	126	~
041	)	084	T	127	DEL
042	*	085	U		

Figure 2 - ASCII character codes. From <https://ee.hawaii.edu/~tep/EE160/Book/chap4/subsection2.1.1.1.html>

Software Flowchart

The following software flowchart is a modified version of that shown in textbook chapter 14.9, and should be used as a general guide for writing the program used in this lab.

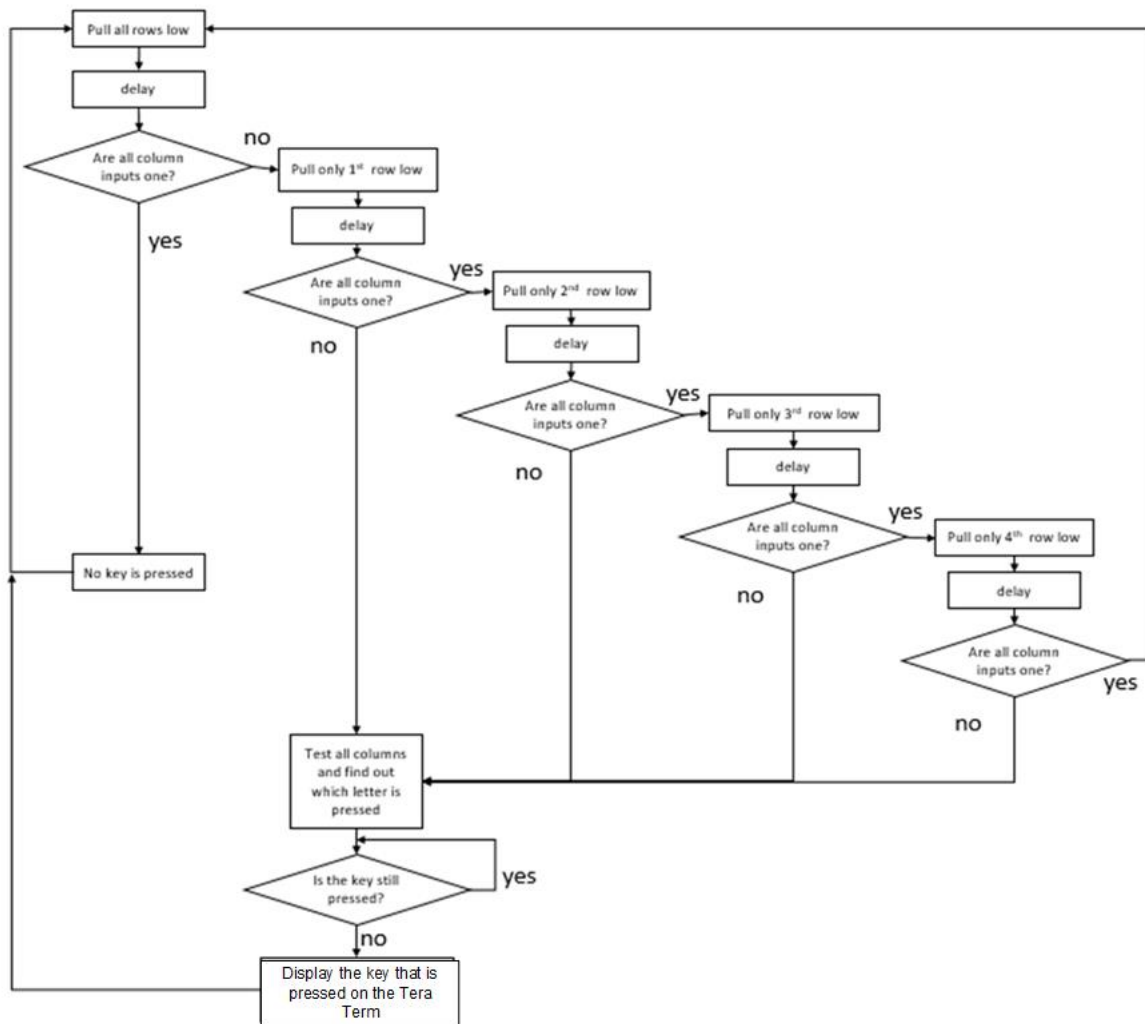


Figure 3 - Software Flowchart for the keypad scanning algorithm

Pre-Lab Register Tables (5 points)

Configure Port C: Pin 0, 1, 2, and 3 as Digital Output

GPIO Mode: Digital Input (00), Digital Output (01), Alternative Function (10), Analog (11)

Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODER	MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]		MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
MASK																																
VALUE																																

GPIOC Mode Register MASK Value = 0x_____ (in HEX)

GPIOC Mode Register Value = 0x_____ (in HEX)

Configure Port B: Pin 1, 2, 3, and 5 as Digital Input

GPIO Mode: Digital Input (00), Digital Output (01), Alternative Function (10), Analog (11)

Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODER	MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]		MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
MASK																																
VALUE																																

GPIOB Mode Register MASK Value = 0x_____ (in HEX)

GPIOB Mode Register Value = 0x_____ (in HEX)

Write to Port C: Pins 0, 1, 2, and 3 connect to the rows of the keypad

Value	ODR	Bit
	Pin 31	31
	Pin 30	30
	Pin 29	29
	Pin 28	28
	Pin 27	27
	Pin 26	26
	Pin 25	25
	Pin 24	24
	Pin 23	23
	Pin 22	22
	Pin 21	21
	Pin 20	20
	Pin 19	19
	Pin 18	18
	Pin 17	17
	Pin 16	16
	Pin 15	15
	Pin 14	14
	Pin 13	13
	Pin 12	12
	Pin 11	11
	Pin 10	10
	Pin 9	9
	Pin 8	8
	Pin 7	7
	Pin 6	6
	Pin 5	5
	Pin 4	4
	Pin 3	3
	Pin 2	2
	Pin 1	1
	Pin 0	0

Value written to PORTC ODR in order to pull down all rows: _____ (in HEX)

Value written to PORTC ODR in order to pull down row 1: _____ (in HEX)

Value written to PORTC ODR in order to pull down row 2: _____ (in HEX)

Value written to PORTC ODR in order to pull down row 3: _____ (in HEX)

Value written to PORTC ODR in order to pull down row 4: _____ (in HEX)

Read from Port B: Pins 1, 2, 3, and 5 connect to the columns of the keypad

Value	ODR	Bit
	Pin 31	31
	Pin 30	30
	Pin 29	29
	Pin 28	28
	Pin 27	27
	Pin 26	26
	Pin 25	25
	Pin 24	24
	Pin 23	23
	Pin 22	22
	Pin 21	21
	Pin 20	20
	Pin 19	19
	Pin 18	18
	Pin 17	17
	Pin 16	16
	Pin 15	15
	Pin 14	14
	Pin 13	13
	Pin 12	12
	Pin 11	11
	Pin 10	10
	Pin 9	9
	Pin 8	8
	Pin 7	7
	Pin 6	6
	Pin 5	5
	Pin 4	4
	Pin 3	3
	Pin 2	2
	Pin 1	1
	Pin 0	0

Mask to check if a button from column 1 has been pressed: _____ (in HEX)

Mask to check if a button from column 2 has been pressed: _____ (in HEX)

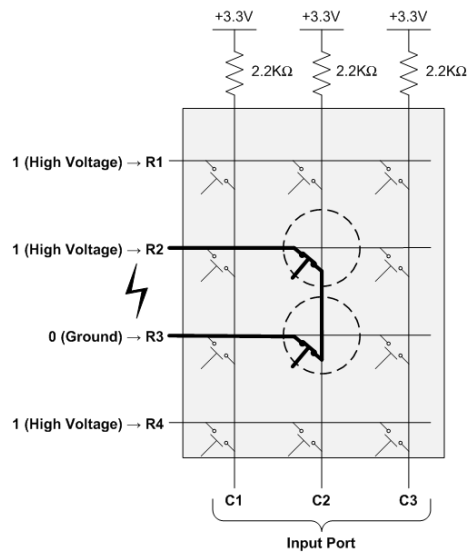
Mask to check if a button from column 3 has been pressed: _____ (in HEX)

Mask to check if a button from column 4 has been pressed: _____ (in HEX)

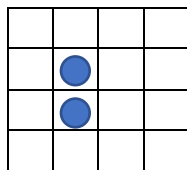
Post-Lab Questions (20 points)

Please include answers to the following questions with your submission of the pre-lab register contents:

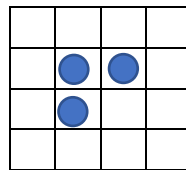
1. How is software debouncing implemented in your program? (3 points)
2. What do we mean when we say that the STM32L4's internal pull-up resistors are too weak for this application? (5 points)
3. When multiple keys are pressed, there could be a short circuit (as shown in the figure). How to configure the output GPIO to avoid this scenario? (7 points)



4. In the following 2 cases, can the scan algorithm correctly detect all keys pressed? If so, how to modify the flowchart (figure 4) of the scan algorithm. If not, explain the reason. (5 points)



Case1



Case2

● Means the keys that are simultaneously pressed.

Participation and Contribution

Please indicate the participation and contribution for each group member using the following table.

Name	Participation and Contribution